

SPECTRA: Stellar Parameter Estimation and Calculation Tools for Research and Analysis

João Paulo Almeida da Silva Matos ^{1,2¶}, Maria Jaqueline Vasconcelos ²,
and Adriano Hoth Cerqueira ²

¹ Department of Physics, São Paulo State University (UNESP), Institute of Biosciences, Humanities and Exact Sciences, São José do Rio Preto, SP, 15054-000, Brazil ² Laboratório de Astrofísica Teórica e Observacional, Universidade Estadual de Santa Cruz (UESC), Ilhéus, Brasil ¶ Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

Determining fundamental stellar parameters — such as mass, age, luminosity, effective temperature (T_{eff}), radius, and distance — is central to observational and theoretical astrophysics (Serenelli et al., 2021). Interior stellar structure (Baraffe et al., 2015), rotation (Vasconcelos et al., 2022), and magnetic field generation (Charbonneau, 2014) are mass- and age-dependent processes. **SPECTRA**¹ (*Stellar Parameter Estimation and Calculation Tools for Research and Analysis*, v1.1.0) is an open-source Python desktop application designed to optimize and automate stellar mass and age determination and diagnostic modeling for stars in clusters, with a particular focus on pre-main-sequence (PMS) and low-mass stars.

The software integrates thread-safe data state management (`spectra.state`), missing data imputation via k -Nearest Neighbors (k -NN) and iterative techniques, 2D Bayesian posterior isochrone fitting (`spectra.bayesian`), machine-learning regression modeling (`spectra.tools`), and interactive diagnostic visualization tools (Baraffe et al., 2015; Serenelli et al., 2021).

Statement of need

In Astrophysics, estimating the mass and age of single stars outside binary systems depends heavily on theoretical stellar evolution models and mass-luminosity or mass-magnitude relations (MLR/MMR) (Benedict et al., 2016; Serenelli et al., 2021). Taking advantage of age and metallicity homogeneity of stars belonging to clusters, isochrone or stellar track fitting on color magnitude and HR diagrams proves to be a powerful tool to obtain simultaneously stellar ages and mass. However, clusters with ages ≤ 100 Myr have a great number of pre-main-sequence stars that lie above the main sequence, exhibiting high observational uncertainties due to differential interstellar extinction, age dispersion, and incomplete data vectors (Herczeg & Hillenbrand, 2015). Differences between theoretical input physics in models (e.g., Siess et al. 2000 vs. BHAC15) also introduce systematic variances (Lebreton et al., 2014).

SPECTRA addresses these challenges by offering a unified, user-friendly desktop application built with `ttkbootstrap`. It enables researchers and educators to:

1. Automatically preprocess, filter, convert linear luminosity units ($L/L_{\odot} \rightarrow \log_{10} L$), and impute incomplete photometric datasets.
2. Perform 2D Bayesian posterior probability inference over theoretical evolutionary grids (Siess et al. 2000; BHAC15) to calculate marginal expected values and $1-\sigma$ uncertainties

¹The development of this software and its underlying research framework were initiated and primarily executed at the Universidade Estadual de Santa Cruz (UESC), with final updates and infrastructure migration supported by the Universidade Estadual Paulista (UNESP).

- 38 for both mass ($M_{\odot}$) and age (Myr) (Baraffe et al., 2015).
39 3. Train, optimize (via Grid Search and K -fold cross-validation), and statistically evaluate
40 machine learning regression algorithms (e.g., Random Forests, Support Vector Regressors,
41 k -NN, Bayesian Ridge) to construct semi-empirical mass-magnitude relations.
42 4. Organize, inspect, and export publication-ready tables (`_final_result_table.csv`) and
43 diagnostic visual figures (Mass Plot, Age Plot, HR Diagram Plot, and 4-panel regression
44 dashboards).

45 State of the field

46 Several computational tools exist within the astronomical ecosystem for stellar parameter
47 estimation and cluster analysis:

- 48 ■ Machine learning tools applied to stellar parameter derivation, such as CARMENES
49 transfer-learning routines (Bello-García et al., 2023), target specific spectral types (M
50 dwarfs) rather than open clusters.
- 51 ■ Empirical relations based on low-mass field binaries (Benedict et al., 2016; Mann et
52 al., 2019) provide benchmarks for main-sequence objects, but lack integrated GUI
53 workflows for cluster-wide missing-data imputation, 2D Bayesian posterior estimation,
54 and pre-main-sequence fits.

55 SPECTRA fills this gap by combining GUI-driven visual diagnostics (e.g., Hertzsprung-Russell
56 and diagnostic residual plots), automated missing-data imputation, 2D Bayesian parameter
57 inference, and a model-ranking framework based on goodness-of-fit metrics ($RMSE$, MAE ,
58 R^2 , and AIC).

59 Software design and workflow

60 SPECTRA is written in Python 3 and relies on core scientific packages including numpy,
61 pandas, scipy, scikit-learn, statsmodels, and ttkbootstrap (Gomes, 2024). The
62 application's architecture decouples GUI control logic from heavy computational backends,
63 providing a modular design structured around thread-safe state management (DataManager in
64 `spectra.state`) and specialized analytical modules.

65 Figure 1 illustrates the high-level execution flow and operational logic of SPECTRA:

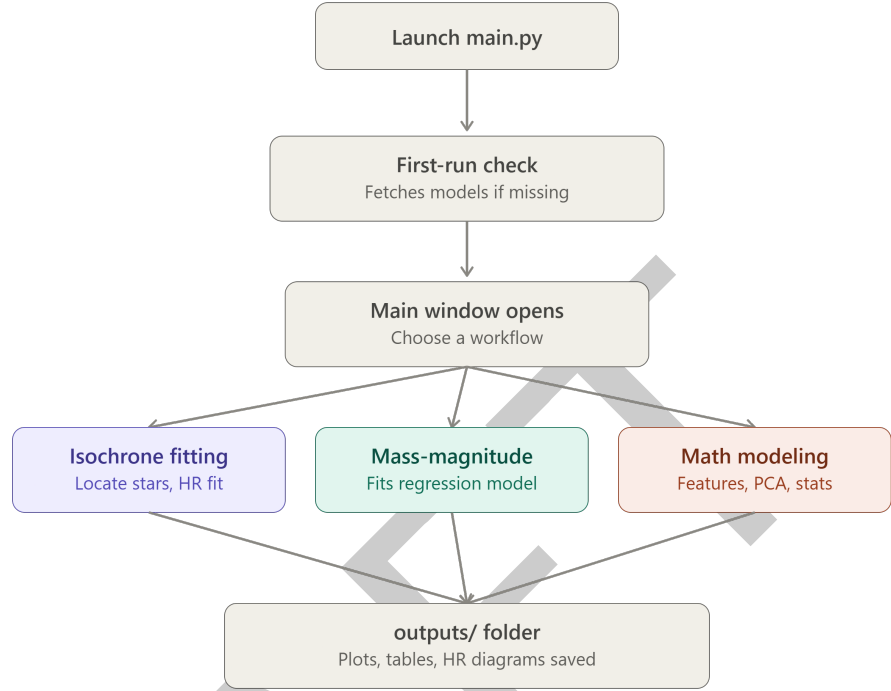


Figure 1: Execution workflow and modular architecture of SPECTRA.

Execution Pipeline & Module Description

1. **Initialization & Environment Setup:** Upon launching `main.py`, SPECTRA performs an automated verification check. If local theoretical evolutionary grids (e.g., Siess et al. 2000; BHAC15) or model weights are missing, the system automatically fetches and caches them to ensure reproducible execution (Baraffe et al., 2015).

2. **Core Analytical Workflows:** From the primary desktop window, users can direct multi-catalog photometry (e.g., *Gaia* DR3, TESS) into three specialized processing modules (Babusiaux et al., 2023; Paegert et al., 2021; Stassun et al., 2019):

- **2D Bayesian Isochrone Fitting Module (`spectra.bayesian`):** Evaluates a full 2D likelihood χ^2 over the HR diagram space:

$$\chi^2 = \left(\frac{T_{\text{eff,obs}} - T_{\text{eff,grid}}}{\sigma_T} \right)^2 + \left(\frac{\log L_{\text{obs}} - \log L_{\text{grid}}}{\sigma_L} \right)^2$$

combined with a Salpeter/Kroupa Initial Mass Function (IMF) prior ($P(M) \propto M^{-1.35}$ for $M \leq 0.5M_{\odot}$, $M^{-2.35}$ for $M > 0.5M_{\odot}$). Marginal posterior moments yield the expected values and physical 1- σ uncertainties for both stellar mass ($\hat{M}, \sigma_M$) and age ($\hat{t}, \sigma_t$).

- **Mass-Magnitude Modeling Module (`spectra.tools`):** Builds semi-empirical relations by training candidate regression algorithms (e.g., k -NN, Random Forests, Support Vector Regressors, Bayesian Linear Regression) over theoretical stellar grids. Hyperparameters are optimized via K -fold cross-validation and Grid Search. Candidate models are evaluated and automatically ranked using a normalized multi-metric score S :

$$S = \hat{\mathcal{M}} \cdot \mathcal{P}^T$$

where $\hat{\mathcal{M}}$ represents the min-max normalized metrics matrix and $\mathcal{P}$ corresponds to a unit vector, assigning equal relative importance to $RMSE$, MAE , $(1 - R^2)$, and AIC .

- **Mathematical Modeling Module:** Provides advanced preprocessing, dataset filtering, missing attribute imputation (e.g., k -NN and iterative stochastic techniques), feature correlation analysis, Principal Component Analysis (PCA), and residual diagnostic checks (homoscedasticity and normality of errors).

3. **Output Artifact Generation:** All generated diagnostics—including annotated HR diagrams, individual mass/age plots (Mass Plot, Age Plot), statistical summary tables (`_final_result_table.csv`), regression report logs, and diagnostic residual plots—are automatically organized and saved in the localized outputs/ repository for direct export into publications.

Research impact & empirical case study

The practical effectiveness of SPECTRA was validated through a comprehensive study of young stellar clusters covering distinct evolutionary stages: Orion Nebula Cluster (ONC, ~ 1 Myr) (Hillenbrand, 1997), Upper Scorpius (~ 5 – 11 Myr) (Pécaut et al., 2012; Rebull et al., 2018), h Persei / NGC 869 (~ 13 Myr), Pleiades (~ 112 Myr) (Dahm, 2015; Gossage et al., 2018; Lodieu et al., 2019), and NGC 2516 (~ 150 Myr).

Methodological Validation

Using a control sample of isolated eclipsing binary stars from Benedict et al. (2016), SPECTRA's k -NN regression model derived from the BHAC15 isochrone grid (Baraffe et al., 2015) achieved high predictive accuracy ($R^2 = 0.96$, $P_{\text{skedasticity}} = 0.12$) with a median residual uncertainty of 3.8%.

Cluster Analysis Results

- **Young PMS Clusters (ONC & Upper Scorpius):** For clusters rich in pre-main-sequence objects with significant age dispersion (Hillenbrand, 1997; Pécaut et al., 2012), 2D Bayesian IsocFit successfully identified low-mass PMS objects and derived simultaneous ages, whereas k -NN regression provided robust mass estimates with less dependency on bolometric corrections (Pécaut & Mamajek, 2013).
- **Well-Defined Main-Sequence Clusters (Pleiades & NGC 2516):** On clusters with established main sequences (Lodieu et al., 2019), k -NN RMM modeling using *Gaia* G -band and V -band absolute magnitudes yielded near-perfect alignment with literature values ($R^2 > 0.99$, $RMSE < 0.01M_{\odot}$), demonstrating superior performance over traditional isochrone grid edge-effects.

Figure 2 illustrates a comparative evaluation of mass estimates derived by SPECTRA for member stars of the Pleiades cluster using both the grid-interpolation method (IsocFit - BHAC15, left panels) and the machine-learning Mass-Magnitude relation (KNN - BHAC15, right panels), benchmarked against literature values from TESS V8.2 (Stassun et al., 2019), (Lodieu et al., 2019), and (Delfosse et al., 2000), (Benedict et al., 2016).

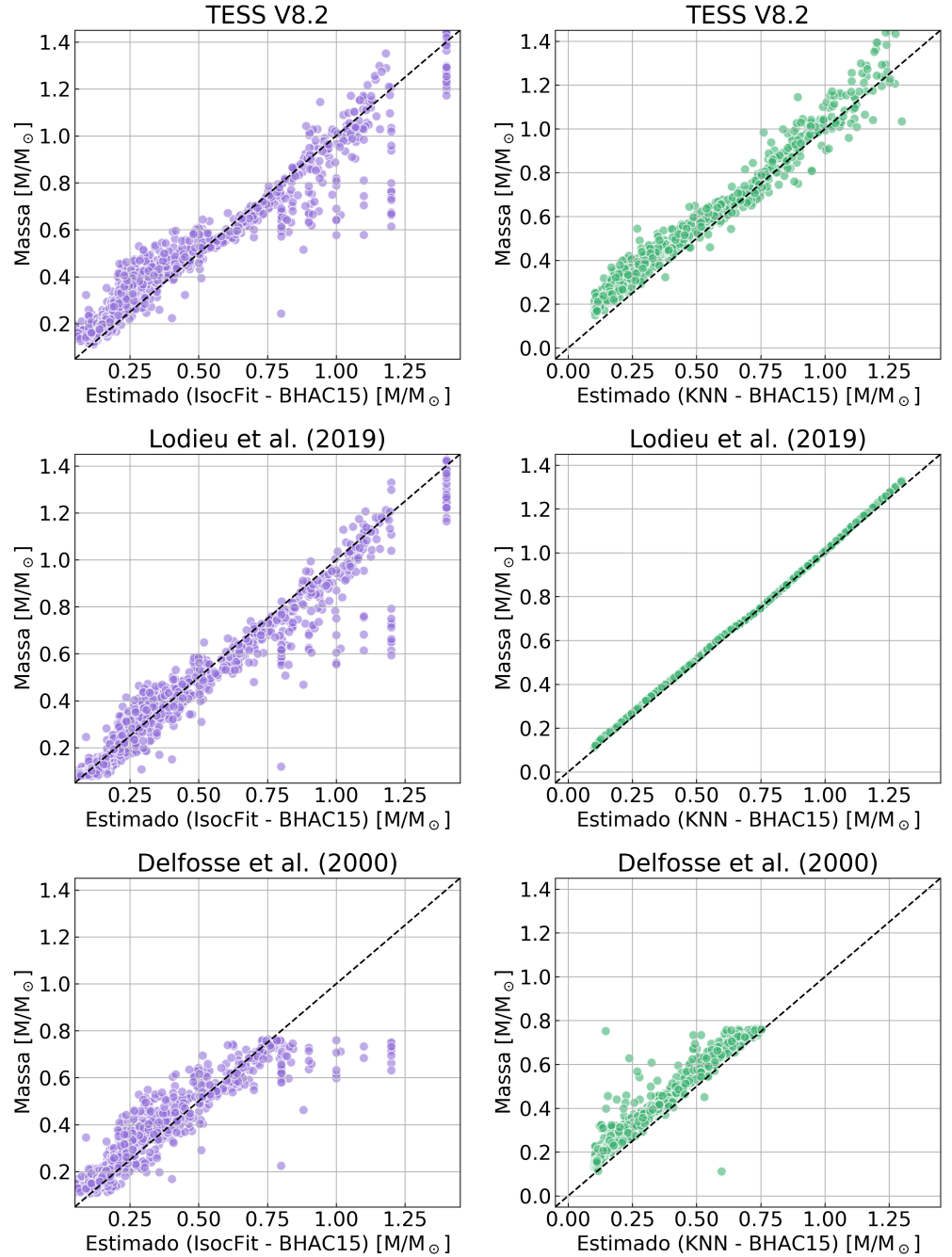


Figure 2: Comparison of stellar mass estimates for the Pleiades cluster obtained via IsocFit (left column) and KNN - BHAC15 (right column) against reference datasets from TESS V8.2, Lodieu et al. (2019), and Delfosse et al. (2000). The dashed line represents the ideal 1:1 identity relation.

As shown in Figure 2, the IsocFit - BHAC15 method exhibits greater dispersion around the 1:1 identity line, particularly for stars with $M > 0.7M_{\odot}$. This behavior is attributed to the convergence of evolutionary tracks near the Zero-Age Main Sequence (ZAMS) at the age of the Pleiades (~ 112 Myr), which creates localized degeneracies during grid minimization (Gomes, 2024). Conversely, the semi-empirical KNN - BHAC15 model demonstrates exceptional predictive stability, showing near-perfect alignment ($R^2 \approx 1.0$) when cross-validated against the photometrically derived masses of Lodieu et al. (2019). When benchmarked against empirical

binary masses from Benedict et al. (2016), the k -NN model reproduces stellar masses with minimal bias up to the applicability threshold of the empirical K -band relation ($M \lesssim 0.75M_{\odot}$) (Gomes, 2024).

A more detailed astrophysical discussion and complete parameter catalogues derived for each cluster will be presented in a forthcoming paper (Gomes et al., in prep.).

By reducing manual preprocessing time and providing rigorous statistical diagnostic reports, SPECTRA supports both observational research in stellar rotation/dynamics and undergraduate/graduate instruction in stellar astrophysics (Gomes, 2024).

AI usage disclosure

No generative AI tools were used in the algorithmic development or code implementation of the SPECTRA package.

Acknowledgements

The primary development, algorithmic design, and empirical cluster validations of SPECTRA were conducted at the Laboratório de Astrofísica Teórica e Observacional (LATO) at the Universidade Estadual de Santa Cruz (UESC), whose financial and structural support during the initial phases of this research was fundamental. The primary author also acknowledges the current institutional support provided by the Universidade Estadual Paulista (UNESP) during the final stages and preparation of this software manuscript, as well as the entire open-source Python scientific community. The work of MJV and AHC is partially funded by the UESC's projects 2023.0003119-74 and 2025.0033125-91.

References

- Babusiaux, C., Fabricius, C., Khanna, S., Muraveva, T., Reylé, C., Spoto, F., Vallenari, A., Luri, X., Arenou, F., Álvarez, M. A., Anders, F., Antoja, T., Balbinot, E., Barache, C., Bauchet, N., Bossini, D., Busonero, D., Cantat-Gaudin, T., Carrasco, J. M., ... Weiler, M. (2023). *Gaia* Data Release 3: Catalogue validation. *Astronomy & Astrophysics*, 674, A32. <https://doi.org/10.1051/0004-6361/202243790>
- Baraffe, I., Homeier, D., Allard, F., & Chabrier, G. (2015). New evolutionary models for pre-main sequence and main sequence low-mass stars down to the hydrogen-burning limit. *Astronomy & Astrophysics*, 577, A42. <https://doi.org/10.1051/0004-6361/201425481>
- Bello-García, A., Passetger, V. M., Ordieres-Meré, J., Schweitzer, A., Caballero, J. A., González-Marcos, A., Ribas, I., Reiners, A., Quirrenbach, A., Amado, P. J., Béjar, V. J. S., Cifuentes, C., Henning, Th., Kaminski, A., Luque, R., Montes, D., Morales, J. C., Pedraz, S., Tabernero, H. M., & Zechmeister, M. (2023). The CARMENES search for exoplanets around M dwarfs: A deep transfer learning method to determine t_{eff} and $[M/H]$ of target stars. *Astronomy & Astrophysics*, 673, A105. <https://doi.org/10.1051/0004-6361/202243934>
- Benedict, G. F., Henry, T. J., Franz, O. G., McArthur, B. E., Wasserman, L. H., Jao, W.-C., Cargile, P. A., Dieterich, S. B., Bradley, A. J., Nelan, E. P., & Whipple, A. L. (2016). THE SOLAR NEIGHBORHOOD. XXXVII. THE MASS-LUMINOSITY RELATION FOR MAIN-SEQUENCE M DWARFS*. *The Astronomical Journal*, 152(5), 141. <https://doi.org/10.3847/0004-6256/152/5/141>
- Charbonneau, P. (2014). Solar Dynamo Theory. 52, 251–290. <https://doi.org/10.1146/annurev-astro-081913-040012>

- 174 Dahm, S. E. (2015). REEXAMINING THE LITHIUM DEPLETION BOUNDARY IN THE
175 PLEIADES AND THE INFERRED AGE OF THE CLUSTER. *The Astrophysical Journal*,
176 813(2), 108. <https://doi.org/10.1088/0004-637X/813/2/108>
- 177 Delfosse, X., Forveille, T., Ségransan, X., Beuzit, J.-L., Udry, S., Mayor, M., & Perrier,
178 C. (2000). Accurate masses of very low mass stars. IV. Improved mass-luminosity
179 relations. *Astronomy & Astrophysics*, 364, 217–224. <https://doi.org/10.48550/arXiv.astro-ph/0010586>
180
- 181 Gomes, J. P. M. D. (2024). *Determinação da massa de estrelas jovens em aglomerados*
182 *estelares: Uma abordagem observacional e estatística* [Trabalho de Conclusão de Curso
183 (Bacharelado em Física)]. Universidade Estadual de Santa Cruz (UESC).
- 184 Gossage, S., Conroy, C., Dotter, A., Choi, J., Rosenfield, P., Cargile, P., & Dolphin, A. (2018).
185 Age Determinations of the Hyades, Praesepe, and Pleiades via MESA Models with Rotation.
186 *The Astrophysical Journal*, 863(1), 67. <https://doi.org/10.3847/1538-4357/aad0a0>
- 187 Herczeg, G. J., & Hillenbrand, L. A. (2015). EMPIRICAL ISOCHRONES FOR LOW MASS
188 STARS IN NEARBY YOUNG ASSOCIATIONS. *The Astrophysical Journal*, 808(1), 23.
189 <https://doi.org/10.1088/0004-637X/808/1/23>
- 190 Hillenbrand, L. A. (1997). On the Stellar Population and Star-Forming History of the Orion
191 Nebula Cluster. *The Astronomical Journal*, 113, 1733. <https://doi.org/10.1086/118389>
- 192 Lebreton, Y., Goupil, M.-J., & Montalbán, J. (2014). Stellar evolutionary models: Current
193 status and challenges. *Astrophysics and Space Science*, 109, 23–38. <https://doi.org/10.1007/s10509-014-1848-1>
194
- 195 Lodieu, N., Pérez-Garrido, A., Smart, R. L., & Silvotti, R. (2019). A 5D view of the
196 α Per, Pleiades, and Praesepe clusters. *Astronomy & Astrophysics*, 628, A66. <https://doi.org/10.1051/0004-6361/201935533>
197
- 198 Mann, A. W., Dupuy, T., Kraus, A. L., Gaidos, E., Ansdell, M., Ireland, M., Rizzuto,
199 A. C., Hung, C.-L., Dittmann, J., Factor, S., Feiden, G., Martinez, R. A., Ruíz-
200 Rodríguez, D., & Chia Thao, P. (2019). How to Constrain Your M Dwarf. II. The
201 Mass–Luminosity–Metallicity Relation from 0.075 to 0.70 Solar Masses. *The Astrophysical*
202 *Journal*, 871(1), 63. <https://doi.org/10.3847/1538-4357/aaf3bc>
- 203 Paegert, M., Stassun, K. G., Collins, K. A., Pepper, J., Torres, G., Jenkins, J., Twicken, J. D.,
204 & Latham, D. W. (2021). *TESS Input Catalog versions 8.1 and 8.2: Phantoms in the 8.0*
205 *Catalog and How to Handle Them*. arXiv. <https://doi.org/10.48550/ARXIV.2108.04778>
- 206 Pecauc, M. J., & Mamajek, E. E. (2013). INTRINSIC COLORS, TEMPERATURES, AND
207 BOLOMETRIC CORRECTIONS OF PRE-MAIN-SEQUENCE STARS. *The Astrophysical*
208 *Journal Supplement Series*, 208(1), 9. <https://doi.org/10.1088/0067-0049/208/1/9>
- 209 Pecauc, M. J., Mamajek, E. E., & Bubar, E. J. (2012). A REVISED AGE FOR UPPER
210 SCORPIUS AND THE STAR FORMATION HISTORY AMONG THE F-TYPE MEMBERS
211 OF THE SCORPIUS-CENTAURUS OB ASSOCIATION. *The Astrophysical Journal*, 746(2),
212 154. <https://doi.org/10.1088/0004-637X/746/2/154>
- 213 Rebull, L. M., Stauffer, J. R., Cody, A. M., Hillenbrand, L. A., David, T. J., & Pinsonneault,
214 M. (2018). Rotation of Low-mass Stars in Upper Scorpius and ρ Ophiuchus with K2. *The*
215 *Astronomical Journal*, 155(5), 196. <https://doi.org/10.3847/1538-3881/aab605>
- 216 Serenelli, A., Weiss, A., Aerts, C., Angelou, G. C., Baroch, D., Bastian, N., Beck, P. G.,
217 Bergemann, M., Bestenlehner, J. M., Czekala, I., Elias-Rosa, N., Escorza, A., Van Eylen,
218 V., Feuillet, D. K., Gandolfi, D., Gieles, M., Girardi, L., Lebreton, Y., Lodieu, N., ... Zwintz,
219 K. (2021). Weighing stars from birth to death: Mass determination methods across the
220 HRD. *The Astronomy and Astrophysics Review*, 29(1), 4. <https://doi.org/10.1007/s00159-021-00132-9>
221

- 222 Stassun, K. G., Oelkers, R. J., Paegert, M., Torres, G., Pepper, J., Lee, N. D., Collins, K.,
223 Latham, D. W., Muirhead, P. S., Chittidi, J., Rojas-Ayala, B., Fleming, S. W., Rose, M.
224 E., Tenenbaum, P., Ting, E. B., Kane, S. R., Barclay, T., Bean, J. L., Brassuer, C. E., ...
225 Winn, J. N. (2019). The Revised TESS Input Catalog and Candidate Target List. *The*
226 *Astronomical Journal*, 158(4), 138. <https://doi.org/10.3847/1538-3881/ab3467>
- 227 Vasconcelos, M. J., Bouvier, J., Gallet, F., & Luz Filho, E. A. (2022). Monte Carlo simulations
228 of evolving rotational distributions of low-mass stars in young open clusters. Testing
229 the influence of initial conditions. *510*(1), 1528–1540. [https://doi.org/10.1093/mnras/](https://doi.org/10.1093/mnras/stab3376)
230 [stab3376](https://doi.org/10.1093/mnras/stab3376)

DRAFT